Enhanced Landmark Detection in Oral and Maxillofacial CT and CBCT Scans: A Multistage Global–Local Integration Approach

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***Abstract*—Cephalometric analysis plays a crucial role in oral clinic, especially in orthodontics and maxillofacial surgery, it is essential for precise and reliable treatment planning and diagnosis. Accurate detection of anatomical landmarks is vital for successful cephalometric analysis. Although significant advancements in deep learning-based landmark detection from medical images have been achieved, especially two-stage coarse-to-fine approaches, the performance remains constrained by the high resolution of images and the consequent limitation of computational resources, particularly with 3-D medical data. In this study, we propose a multistage approach to precisely detect landmarks from 3-D computed tomography (CT) and cone beam CT (CBCT) images. First, we utilize an alignment network to identify the region of interest and reduce redundant information. Next, we introduce a novel coarse-to-fine detection framework employing 3-D U-Net model with heatmap regression as backbones. Specifically, we propose a multiscale global information extraction module (MSGIEM) that combines features from multiple layers with varying resolutions and semantics to capture effective global information. Additionally, a landmark attention module (LAM) is introduced to distill relevant global features for individual landmarks, facilitating their integration with local visual patches to enhance overall detection performance. Extensive experiments on in-house CT and CBCT datasets, as well as a public dataset, demonstrate that the proposed method significantly outperforms state-of-the-art approaches, with a notable reduction on the mean radial error (MRE) by 0.14 and 0.04 mm, and an improvement on the successful detection rate (SDR) within 2 mm by 3.98% and 0.23% on in-house CT and CBCT datasets, respectively. Furthermore, our discussion on the detection of common landmarks across CT and CBCT datasets underscores its potential clinical applications. The code is available at https://github.com/HLZ-CODE/Maxillofacial-Landmark**

***Index Terms*—Cephalometric landmark detection, computed tomography (CT) and cone beam CT (CBCT), global–local integration, heatmap regression.**

Received 19 December 2024; revised 18 March 2025; accepted 2 May 2025. Date of publication 19 May 2025; date of current version 11 June 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62376176 and in part by the Science and Technology Department of Sichuan Province under Grant 2023YFG0272. The Associate Editor coordinating the review process was Dr. Bardia Yousefi. *(Corresponding author: Jixiang Guo.)*

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Digital Object Identifier 10.1109/TIM.2025.3571083

I. Introduction

Cephalometric analysis is a crucial mean in oral clinical treatment to evaluate the dental and skeletal relationship of a human skull. It is widely used by the dentists, orthodontists, and oral and maxillofacial surgeons as a treatment designation and planning tool [1]. During the cephalometric analysis, the anatomical landmark detection is the first and most important step, which provides the bony and soft tissue landmarks for subsequent study of patient-specific proportions and measurements of angles. In recent years, a lot of research has been done on the deep learning-based landmarks detection from 2-D lateral radiographs. This saves doctors away from tedious manual workload and improves the efficiency of orthodontic treatment. However, compared to 2-D images, the 3-D images such as computed tomography (CT) and cone beam CT (CBCT) scan can provide more plentiful and valuable structural information to assist treatment planning and evaluation for the patient with deformities. Specifically, the 3-D spatial relationship analysis has significant role in the planning of oral and maxillofacial surgery. In this work, we aim to present an accurate and effective framework for landmarks detection from 3-D CT and CBCT images.

To automatically detect landmarks, the existing methods are mainly divided into three categories: knowledge and graph-based methods [2], [3], machine learning methods, and deep learning methods. Knowledge and graph-based methods use graph images with annotated landmarks to vote and determine the position of landmarks in the image. Although these methods have achieved some degree of success, they are highly time-consuming and labor-intensive. Traditional machine learning methods are usually based on classification or regression. Classification-based methods detect the presence of landmarks in slices, patches, or voxels, and use hard thresholds to determine the classification results [4], [5]. Regression-based methods estimate displacement vectors from slices, patches, or voxels to landmarks [6], [7]. Unfortunately, the results of these methods are usually inadequate for clinical application. With the advent of deep learning, researchers have gradually applied deep learning algorithms to cephalometric analysis. The existing deep learning methods are generally divided into coordinate regression methods and heatmap regression methods. The coordinate regression methods directly learn the mapping from images to coordinates, while the heatmap regression methods learn the mapping from images to Gaussian heatmaps of landmark, and then extract landmarks from the heatmap. Moreover, the deep learning-based methods can be structurally divided into single-stage methods and multistage methods. The single-stage methods directly locate the coordinates of landmarks from the image through a single network. However, directly identifying landmarks from complete high-resolution images leads to higher requirements on computing resources and affects detection accuracy due to the large amount of redundant information. Fortunately, multistage methods cascaded from coarse to fine have been shown to enhance performance for landmark detection. These methods have consequently become the mainstream methods in deep learning-based landmark detection approaches.

For multistage methods, most studies adopted two-stage framework. In the first/global stage, downsampled low-resolution images are typically used for coarse landmark detection. In the second/local stage, based on the detection results from the global stage, high-resolution patches are cropped from the original images for precise landmark detection. On this basis, many researchers have further proposed to integrate the information of global stage into the local stage to enhance detection accuracy. Lee et al. [8] used a two-stage coordinate regression method for landmark detection. They extract global information from the lowest layer of the global stage, which is then combined with patches through an attention mechanism to achieve more accurate local detection performance. Chen et al. [9] proposed a two-stage network named structure-aware long and short-term memory network (SA-LSTM) that combines global structural information for landmark detection. They extract the lowest level features of the global stage as global information, and integrate them into the local stage through graph attention. The CephalFormer proposed by Jiang et al. [10] takes the Gaussian heatmap of the estimated landmarks in the global stage as a global structural dependency and inputs it together with the cropped patches into the local stage, here the fusion is performed through transformer. Given the above, although these methods utilize the structural information of the global stage, they simply take the feature map of a certain layer of the global network as complete global information. In fact, the global features at different semantic levels may have different impacts on the detection of local stage. Moreover, different landmarks focus on different global information. These methods do not selectively extract specific global information for different landmarks. Although SA-LSTM slices the regions near different landmarks on the global feature map to represent the unique global information of different landmarks, it actually discards the effective information at long distances. Thus, how to better utilize effective global information to enhance landmark detection deserves further investigation.

In this work, we propose a three-stage framework to accurately detect landmarks from CT and CBCT images. In the first stage, we use an alignment network to crop out the region of interest, i.e., the oral and maxillofacial region. In the second stage, the 3-D U-Net [11] based on heatmap regression is adopted for coarse landmark detection. Inspired by [12], to obtain the effective global information, we propose a novel multiscale global information extraction module (MSGIEM) to combine multiple layers' features with different resolutions and semantics in the 3-D U-Net encoder. Furthermore, we propose a landmark attention module (LAM) to capture the corresponding global information for different landmarks. In the third stage, we input the cropped patch of each landmark and its corresponding global information into another 3-D U-Net for fine-scale landmark detection.

The main contributions of this article are summarized as follows.

1) We designed a novel method for integrating features from the global stage into the local stage of detection, and treated both stages as a unified framework for training.
2) We proposed an MSGIEM which combines global features of different resolutions and semantics to effectively extract global feature information.
3) We proposed an LAM to capture corresponding global information on features for different categories of landmarks, thereby more effectively integrating global information into local stage detection.
4) We establish a comprehensive multimodal dataset to rigorously evaluate the method's robustness in diverse clinical scenarios. The proposed method reduces the MRE by 0.14 and 0.04 mm and improves the SDR within 2 mm by 3.98% and 0.23% compared to state-of-the-art methods on in-house CT and CBCT datasets and outperforms the SOTA methods on a public dataset.

II. Related Works

Since the release of the cephalometric analysis challenge by the IEEE Signal Processing Society (ISBI) in 2014, a wide array of automated landmark detection methods has been proposed. Initially, machine learning techniques were predominantly employed; however, their accuracy often fell short of the stringent requirements for clinical applications. The advent of deep learning marked a significant turning point, as these methods demonstrated markedly superior performance in landmark detection tasks. Notably, Lee et al. [13] pioneered the application of deep learning for automatic cephalometric analysis. They developed 38 regressors to predict the *x*- and *y*-coordinates of 19 landmarks on 2-D lateral radiographs, achieving promising results. Deep learning-based methods have consistently demonstrated superior performance compared to conventional approaches including knowledge-based systems, graph-based techniques, and shallow learning algorithms, typically achieving errors within the clinically acceptable 2 mm range [14]. These advancements have led to breakthroughs in accuracy, generalization, and detection efficiency, with some methods already being integrated into clinical practice. Existing deep learning-based methods can be broadly categorized into five groups, including supervised approaches such as coordinate regression, heatmap regression, object detection, and cascaded multistage methods, as well as reinforcement learning-based techniques.

TABLE I
COMPARISON WITH STATE-OF-THE-ART METHODS ON CT AND CBCT DATASET BY EVALUATION METRICS OF PARAMS, TIMES, MRE, SD, AND SDR IN FOUR TARGET RADII

Dataset	Method	Params (M)	Times (ms)	MRE±SD (mm)	SDR (%)			
					2 mm	2.5 mm	3 mm	4 mm
CT	PWA [8]	33.29	5142	3.70±1.13	25.64	34.56	50.19	69.45
	VNet [39]	45.62	3359	2.49±0.50	43.98	59.67	71.64	86.52
	SA-LSTM [9]	15.33	8296	2.36±0.61	48.87	66.79	78.82	89.65
	SCN [18]	4.08	3662	2.16±0.49	55.95	69.88	79.46	91.93
	LA-GCN [40]	65.9	11536	1.73±0.51	70.07	84.37	90.93	92.35
	DRM [27]	8.15	10354	1.67±0.48	73.62	84.44	88.91	90.41
	ABRM [26]	44.93	1370	1.54±0.49	77.62	86.23	90.11	94.39
	Proposed Method	12.23	12654	**1.40±0.47**	**81.60**	**88.15**	**92.19**	**95.78**
CBCT	PWA [8]	33.29	4957	3.30±0.88	27.65	39.15	54.76	73.02
	SA-LSTM [9]	15.33	7954	2.07±0.49	70.21	80.59	84.32	89.75
	SCN [18]	4.07	3518	1.95±0.55	73.22	82.10	86.14	90.31
	VNet [39]	45.60	1881	1.85±0.30	69.72	81.02	86.14	91.12
	LA-GCN [40]	65.9	10495	1.35±0.35	83.47	89.42	91.92	95.62
	DRM [27]	8.14	10178	1.33±0.40	82.01	88.49	91.80	96.69
	ABRM [26]	44.93	1322	1.24±**0.28**	86.94	**93.25**	**95.77**	**98.94**
	Proposed Method	12.23	9142	**1.20**±0.32	**87.17**	92.72	94.18	96.43

TABLE II
ABLATION EXPERIMENTS ON CT DATASET BY EVALUATION METRICS OF PARAMS, TIMES, MRE, SD, AND SDR IN FOUR TARGET RADII

Config	Params (M)	Times (ms)	Stage	MRE±SD (mm)	SDR (%)			
					2 mm	2.5 mm	3 mm	4 mm
Base	11.96	10299	Global	2.25±0.54	52.25	67.31	77.12	90.19
			Final	1.62±0.47	78.53	86.49	91.75	95.09
Base + GIEM	12.12	10311	Global	2.21±0.53	52.63	66.94	78.19	91.00
			Final	1.57±0.49	79.84	87.08	91.12	95.27
Base + MSGIEM	12.22	11417	Global	2.33±0.51	49.19	65.56	76.18	88.81
			Final	1.49±0.48	79.14	87.02	92.06	95.78
Base + MSGIEM + LAM	12.23	12654	Global	2.28±**0.45**	48.65	64.97	77.32	92.06
			Final	**1.40**±0.47	**81.60**	**88.15**	**92.19**	**95.78**

In this section, we provide a comprehensive overview of recent advancements in deep learning-based cephalometric analysis methods.

A. Coordinate Regression Methods

Early research focused on coordinate regression methods, which directly modeled the mapping of images to coordinates. Subsequent research also involved estimating displacement vectors or offset maps from slices, patches, and voxels to landmarks to obtain landmark coordinates. Julia et al. [15] combined offset regression and classification probability for landmark detection. First, the image was divided into multiple blocks. Then, each block simultaneously estimated the offset and correlation probability from the patch to the landmark. Palazzo et al. [16] introduced a method for predicting 3-D coordinates through a sequential approach. Their framework begins by identifying the sagittal plane (x-coordinate) using 3-D convolutional operations. Subsequently, the volume is sliced along the y- and z-axes, and the resulting 2-D slices are processed independently through two distinct dense layers. The features extracted from these orthogonal directions are then fused, and the final y- and z-coordinates are regressed via a fully connected layer. Lu et al. [17] proposed the utilization of a local attention mechanism at the output of U-Net to extract heatmaps and features of landmark and employ a convolutional graph network, which takes the features as input and determines the presence of landmarks through binary classification, demonstrating enhanced robustness. However, the improvement in MRE was marginal, with a reduction of only 0.03 mm. To sum up, although these methods are capable of directly predicting landmark coordinates, the mapping from image data to spatial coordinates exhibits significant nonlinearity. This poses challenges for model learning, ultimately limiting the detection accuracy. Consequently, research in this area has been relatively scarce over the past two years.

B. Heatmap Regression Methods

The Gaussian heatmap regression method [18] is a widely adopted approach in landmark detection. Its core concept is transforming landmark coordinates into Gaussian-distributed heatmaps, which serve as supervisory signals for training networks. By regressing probability distributions that approximate ground-truth heatmaps, this method enables landmark detection. Specifically, the ground-truth coordinates of each landmark are encoded as a 2-D/3-D Gaussian heatmap, centered at the landmark's coordinates, with the peak probability at the center gradually decaying outward. The network is trained by minimizing the loss between the predicted and ground-truth heatmaps. During inference, the final coordinates are obtained either by identifying the pixel with the maximum value in the predicted heatmap or other similar techniques. By leveraging the spatially continuous representation of heatmaps, this approach enhances robustness to minor positional variations and achieves high accuracy in practical applications. As a result, Gaussian heatmap regression has emerged as a dominant research direction in landmark detection over the years.

Payer et al. [18] proposed the spatial configuration network (SCN), which transforms the coordinate regression problem into a pixel classification problem. The SCN eliminates ambiguous landmarks in the heatmap which solves the problem of landmark misclassification. Zhao et al. [19] applied a two-stage SCN network to detect landmarks of knee and spine in 2022, verifying the landmark detection ability of SCN network on different datasets. Stern et al. [20] proposed a method based on heatmap regression in 2021, which outputs foreground and background heatmaps instead of outputting a heatmap for each landmark. This method can be applied to landmark detection tasks with variable numbers of landmarks. Zhou et al. [21] considered and improved the heatmap regression method by using the variance of the Gaussian heatmap as a learnable parameter for training. It adaptively learned the coordinates and range information of different landmarks, and achieved better detection results for landmarks located in structural areas with different size. Li et al. [22] proposed a feature decoupling and gated recalibration network in 2024 to improve the regression of heatmaps. It enhanced the features of target landmarks in heatmaps and eliminated the features of ambiguous landmarks. Thus, it can more accurately locate target landmarks. Lu et al. [23] combined transformer with SCN and replaced the partial deep level structure of the network with transformer structure, achieving better detection results.

In summary, compared with coordinate regression method, the heatmap regression method has a large output feature map and strong spatial generalization ability, and it usually has higher accuracy. However, the heatmap regression method has a large memory consumption, slow training, and inference speed, and usually needs to reduce image resolution for model training which inevitably result in information loss.

C. Object Detection Methods

Object detection is a widely used technique for identifying and localizing targets within images, typically achieved through the regression and classification of predefined bounding boxes. In recent years, researchers have adapted object detection methods for landmark detection by regarding the center of the detected bounding box as the landmark coordinates. For instance, Chen et al. [24] employed Faster R-CNN to localize coarse landmarks in 3-D cephalometric images, followed by refinement using a multiscale U-Net architecture. This approach effectively addresses the challenge of varying numbers of landmarks across different patients. Lang et al. [25] introduced a novel object detection framework based on Mask R-CNN for landmark localization. More recently, He et al. [26] proposed a multiscale anchor ball regression method, where the offset vectors and categories of landmarks are regressed within each anchor ball. The final landmark coordinates are derived by aggregating predictions from multiple anchor balls.

The primary advantage of these methods lies in their ability to capture the diverse feature ranges of landmarks through bounding boxes, while simultaneously enabling classification to handle the variability in the number of landmarks. However, object detection methods primarily focus on the feature range of landmarks, often resulting in limited accuracy in pinpointing the precise center of the features. As a result, in current research, object detection techniques are predominantly employed in the initial stage of multistage frameworks to achieve coarse localization of landmark regions, which are then refined in subsequent stages.

D. Cascaded Multistage Methods

Due to the high resolution of medical images and the limitations of computational resources, many studies have tried to design multistage detection frameworks that operate from coarse to fine. Typically, in the first/global stage, low-resolution detection is performed to obtain coarse landmark estimates, which are subsequently refined in the second/local stage using high-resolution patches. Here, the patches are cropped from the original high-resolution images. However, a common limitation of these methods is that landmarks may fall outside the cropped regions. To address this issue, Zhong et al. [27] proposed an expansion search method in 2019, which extends the detection range in the local stage by cropping multiple overlapping patches. This approach effectively mitigates the problem of landmarks being excluded from the cropped regions. Nishimoto et al. [28] introduced a three-stage detection model, where ResNet is employed at each stage to estimate landmark coordinates. Cropping is performed between adjacent stages to progressively refine the detection results. Similarly, Zeng et al. [29] proposed a multistage detection network, which incorporates an alignment network in the initial stage to focus on the region of interest and eliminate redundant information, thereby improving overall detection performance. Liu et al. [30] explored a multitask learning approach, combining bone segmentation with landmark detection. They demonstrated that a model pretrained for segmentation tasks exhibits enhanced landmark detection capabilities, highlighting the synergy between these tasks. Dot et al. [31] observed that landmarks located in different anatomical and functional regions exhibit distinct feature characteristics, while those within the same region share similar features. Based on this insight, they proposed a two-stage SCN. In the second stage, landmarks are divided into five groups based on their anatomical regions, and each group is processed separately by dedicated SCN models for landmark refinement. Wang et al. [32] employed a two-stage HR-Net based on heatmap regression for landmark localization, where the second stage leverages low-scale features to further refine the landmark location. However, this method requires exceptionally high input resolutions for both stages, resulting in significant computational complexity.

To sum up, cascaded multistage methods detect landmarks in a coarse-to-fine manner, progressively refining landmark coordinates to achieve high detection accuracy. However, a significant limitation of these methods is the potential loss of information during the cropping operations between stages. To address this issue, several studies have attempted to integrate global-stage information into the local stage to enhance detection performance. For instance, Lee et al. [8] proposed a two-stage network in 2022 that employs patch-level attention to fuse encoded features from both images and patches, achieving remarkable detection accuracy. Similarly, Chen et al. [9] introduced a two-stage network in 2022 that incorporates global structural information into the second stage to assist landmark detection. By embedding this global information, they progressively optimized landmark coordinates using long short-term memory (LSTM). Jiang et al. [10] further advanced this approach by embedding global structural constraints and visual features of patch into a unified feature space, enabling effective information exchange and achieving precise cephalometric landmark detection. Despite these advancements, existing methods primarily extract global structural constraints from features at a single semantic level, which often results in incomplete and inaccurate global information. Additionally, these approaches uniformly integrate the extracted global information into the local detection network across all patches, neglecting the fact that the relevance of global information varies significantly for different landmarks.

E. Reinforcement Learning Methods

In recent years, the rapid advancement of reinforcement learning has inspired its application in landmark detection. Ghesu et al. [33] proposed a novel approach that formulates landmark detection as a path-planning problem, where an agent explores the optimal path from an initial position to the target landmark. Ghesu et al. [34] further enhanced this method by incorporating multiscale image analysis, significantly improving detection accuracy. However, a notable limitation of this approach is its inability to detect multiple landmarks simultaneously. To address this issue, Vlontzos et al. [35] proposed a multiagent reinforcement learning framework in the same year, enabling the joint detection of multiple landmarks. Despite these advancements, reinforcement learning-based methods face significant challenges in handling multiple landmarks concurrently. Moreover, as the number of landmarks increases, the complexity of the search space grows simultaneously, leading to a substantial decline in detection accuracy. Consequently, research on reinforcement learning for landmark detection remains limited, and further exploration is needed to overcome these inherent limitations.

III. METHODOLOGY

Given CT or CBCT scans with an average resolution of $506 \times 646 \times 239$ for CT and $561 \times 403 \times 143$ for CBCT, our method aims to automatically detect N anatomical landmarks $X - [x^1, y^1, z^1, \ldots, x^N, y^N, z^N] \in R^{3\times N}$ from 3-D images using a three-stage method, where $x^i, y^i, z^i \in R^3$ represents the coordinates of the ith landmark.

The overall workflow of our method is illustrated in Fig. 1. Our framework consists of three sequential stages designed for hierarchical detection of oral and maxillofacial landmarks. In the first stage (alignment stage), a 3-D alignment network is utilized to detect and adaptively crop the oral and maxillofacial region-of-interest (ROI) from the original volumetric image. This preprocessing step addresses two key challenges: 1) eliminating redundant background voxels and providing a high-resolution downsampled global image for the subsequent global stage, thereby enhancing detection accuracy and 2) establishing an adaptive cropping mechanism that precisely defines the region of interest, ensuring comprehensive landmark inclusion.

The subsequent two stages established the coarse-to-fine detection framework. In the second stage (global stage), the downsampled image ($128 \times 128 \times 64$) is input to a 3-D U-Net [11] backbone to generate global heatmaps, from which coarse landmark coordinates are derived using a voting mechanism. In the third stage (local stage), high-resolution patches ($32 \times 32 \times 16$) centered on each coarse landmark are cropped from original images to refine precision. These patches, concatenated with contextual features from the global stage, are fed into another 3-D U-Net to predict refined heatmaps through voxelwise classification. Notably, both detection stages employ heatmap regression as their core learning strategy, leveraging the 3-D U-Net architecture as the backbone. In Sections III-A–III-D, we provide a detailed exposition of these three stages and the training process.

A. Alignment Net

As shown in the top-left input image in Fig. 1, CT and CBCT images inherently contain a significant amount of redundant background voxels, which provide minimal utility for cephalometric landmark detection. During the global detection stage, directly downsampling the entire images would lead to reduced resolution of the oral and maxillofacial regions. By first removing redundant background voxels before downsampling, the detection process can focus on higher resolution representations of the oral and maxillofacial regions, thereby enhancing detection performance. Existing methods typically eliminate redundancy through direct cropping; however, this approach risks inadvertently excluding landmarks from the cropped region. To address this limitation, we employ a 3-D alignment network [29] to identify an alignment box that fully encompasses the cephalometric region. The region of interest (ROI) is then cropped from the original image based on this alignment box and subsequently downsampled and fed into the coarse-to-fine landmark detection framework.

B. Coarse-to-Fine Landmark Detection Framework

Although the alignment network effectively removes redundant regions of the input images, the resolution of the remaining ROI is still excessively high for landmark detection, posing a risk of GPU memory overflow. To mitigate this

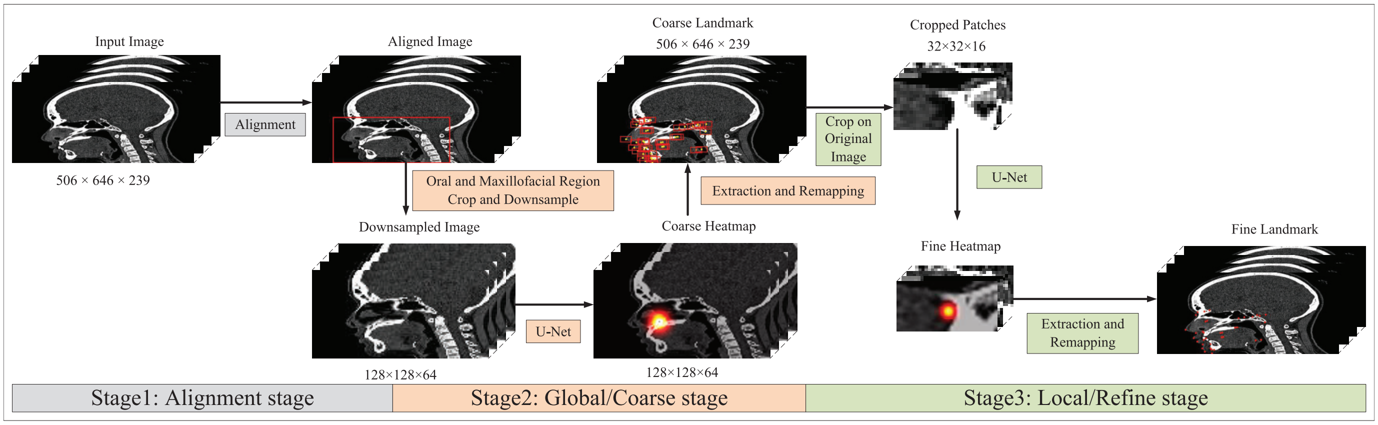

Fig. 1. Workflow of the proposed method. The pipeline consists of three stages. In the first stage, the complete CT or CBCT images are input into an alignment network to focus the images on the oral and maxillofacial region. In the second stage, global landmark detection is performed, while the third stage refines the landmarks on patches.

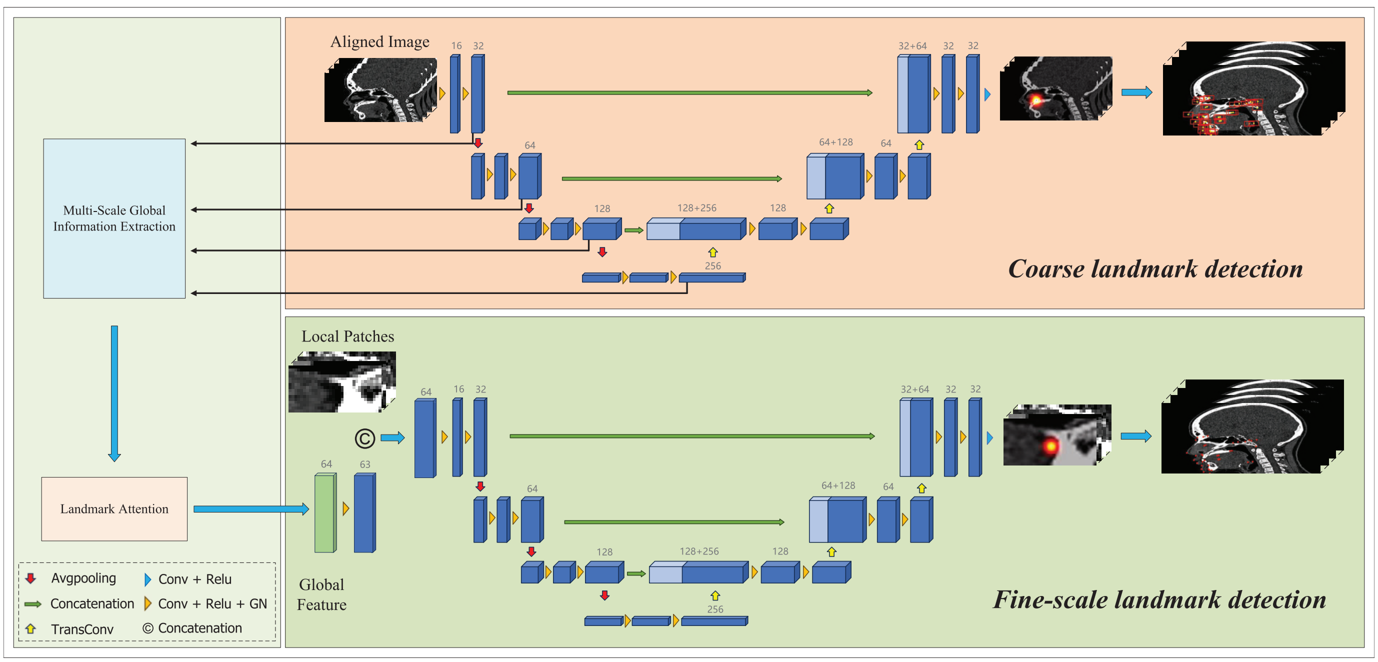

Fig. 2. Illustration of our coarse-to-fine landmark detection framework. The global stage reduces the resolution of the high-resolution oral and maxillofacial images for coarse landmark detection. In the local stage, fine-grained landmark detection is performed by cropping the regions centered on each coarse landmark. The number above the block in the figure represents the number of channels in the feature map.

issue, we propose a coarse-to-fine framework that progressively refines landmark detection, as illustrated in Fig. 2. This framework comprises two stages, both leveraging a 3-D U-Net [11] architecture based on heatmap regression. We adopt a four-level 3-D U-Net architecture. In the encoder path, four hierarchical levels extract features through convolutional operations with channel dimensions of 32, 64, 128, and 256, respectively. The final decoder path progressively restores spatial resolution through upsampling operations, where skip connections with corresponding encoder features are established at each layer to ultimately generate Gaussian heatmaps matching the original input scales. In the global stage, the downsampled oral and maxillofacial region is processed to generate heatmaps for all target landmarks, from which coarse landmark coordinates are derived. Subsequently, high-resolution patches centered on these coarse landmarks are cropped from the original images for the local stage. Furthermore, we extract global contextual information relevant to each landmark from the global stage using our proposed global–local integration strategy. Finally, by concatenating these high-resolution patches with their corresponding global contextual information and feeding them into the network, the framework regenerates heatmaps for all landmarks, enabling the extraction of refined coordinates with enhanced precision.

Guided by empirical evidence in prior studies [36], and considering the experimental environment, we systematically downsample all images to the resolution of $128 \times 128 \times 64$ for input in the global stage. For an input image $I \in R^{128\times128\times64}$ with the corresponding landmarks set X, the Gaussian heatmaps H of all landmarks are directly generated

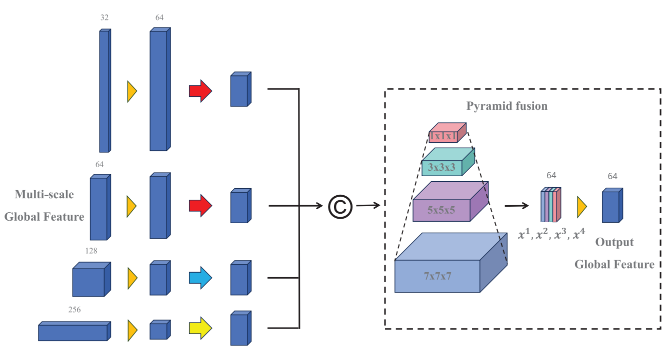

Fig. 3. MSGIEM to obtain global feature of different semantics. The arrows have the same meaning as those in Fig. 2, and the four colors in the pyramid fusion represent four convolution kernels of different sizes.

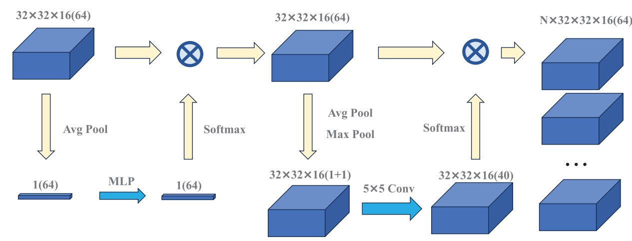

Fig. 4. Attention module for extracting global information for different landmarks.

from the coordinates, where the heatmap of the ith landmark is H^i and $H^i \in R^{128\times128\times64}$. To model the mapping of $I \rightarrow H$, we employ a 3-D U-Net architecture. The input image I is fed into the architecture to generate N predicted heatmaps $\hat{H}$. Then, a voting mechanism is applied to select the top k points with the highest values in $\hat{H}^i$, and the arithmetic mean of these coordinates is designated as the coarse coordinates of the landmark, as represented in the following equations, where r_{focus} represents the focus radius of the heatmap H:

$$X^i = 1/k \times \sum \text{argmax}_{1,2,\ldots,k}\left(\hat{H}^i\right) \tag{1}$$

$$k = \pi r_{\text{focus}}^2. \tag{2}$$

Following the coarse detection in the global stage, a fine-scale adjustment is performed in the local stage. Initially, high-resolution patches centered on the coarse landmarks are cropped from the original images. Similar to work [36], we crop the patches with the resolution of $32 \times 32 \times 16$. The patch for the ith landmark is represented as $I^i_{\text{local}} \in R^{32\times32\times16}$. The corresponding landmarks $X_{\text{local}} - [x^i_{\text{local}}, y^i_{\text{local}}, z^i_{\text{local}}] \in R^3$ are calculated within the patch coordinate system. Notably, the Gaussian heatmap is employed for voxel-level classification. Consequently, the 3-D U-Net models $I_{\text{local}} \rightarrow H_{\text{local}}$ which generate N predicted heatmaps $\hat{H}_{\text{local}}$ for classification. $H^i_{\text{local}} \in R^{32\times32\times16}$ denotes the local Gaussian heatmap of the ith landmark. Finally, the fine coordinates of the ith landmark are extracted from the predicted heatmap $\hat{H}^i_{\text{local}}$ and then mapped back to the coordinate system of the original image.

C. Global–Local Integration

For coarse-to-fine landmark detection, the cropping operation in the local stage generally results in the significant loss of long-distance information. To address this limitation, we propose a novel approach that selectively integrates global-stage information into the local stage. As illustrated in Fig. 3, we design an MSGIEM to capture global information at varying semantic levels. Specifically, we extract the final-layer feature maps from each resolution level of the global-stage U-Net encoder as inputs to the MSGIEM. These feature maps, characterized by varying resolutions and channel dimensions, contain multilevel global anatomical structural information. Within the MSGIEM, convolution operations are first applied to unify the channel number to 64. Subsequently, upsampling and pooling operations are employed to standardize the resolution to $32 \times 32 \times 16$. The four processed feature maps are then concatenated and fused using a pyramid convolution module [37], facilitating the aggregation of global information across multiple semantic levels. The mathematical expression of the process is represented in the following equations:

$$\begin{aligned} F' &= \mathbb{C}\left(\text{AvgPool}\left(\text{Conv}_{1\times1}\left(F^1\right)\right); \text{AvgPool}\left(\text{Conv}_{1\times1}\left(F^2\right)\right)\right) \\ &\quad \text{Conv}_{1\times1}\left(F^3\right); \text{TransConv}\left(\text{Conv}_{1\times1}\left(F^4\right)\right) \end{aligned} \tag{3}$$

$$\begin{aligned} x^i &= \text{ReLU}\left(BN\left(\text{Conv}_{d_i\times d_i}\left(F'\right)\right)\right) \\ F_{ms} &= \text{Conv}_{1\times1}\left(\mathbb{C}\left(x^1, x^2, x^3, x^4\right)\right) \end{aligned} \tag{4}$$

where F' denotes the input feature maps. $\mathbb{C}(\cdot)$ and $\text{Conv}_{1\times1}(\cdot)$ denote the channel concatenation operation and the 1×1 convolution used to reduce the number of channels. $\text{AvgPool}(\cdot)$ denotes the average pooling operation, and $\text{TransConv}(\cdot)$ is a transposed convolution applied for upsampling. Equation (4) is a pyramid fusion module, where $\text{ReLU}(\cdot)$ is the rectified linear unit activation function, $BN(\cdot)$ refers to batch normalization, and $\text{Conv}_{d_i\times d_i}(\cdot)$ indicates a convolution with a kernel size of $d_i \times d_i$, which varies to capture features across different scales. The multiscale global feature map F_{ms} is obtained by concatenating the outputs from different branches x^1, x^2, x^3, x^4 and applying a 1×1 convolution.

Furthermore, acknowledging that different landmarks emphasize distinct global features, we introduce an LAM tailored to generate landmark-specific global features. As illustrated in Fig. 4, a channel attention module [38] is leveraged to amplify the semantic focus of each landmark. Subsequently, the spatial attention module is utilized, where its weight branch generates N independent weights through a multiscale perceptron. These weights reflect the different attention of N landmarks toward global features. They are then multiplied with the global feature map to generate N independent global feature maps, each corresponding to the specific interests of different landmarks. In the local stage, N cropped patches are input into the local network for refined detection of landmarks. In order to fuse global features, the patches and its corresponding landmark global feature F_g are merged and input into the local network. The entire process is expressed in the following equations:

$$\begin{aligned} W_c &= \text{Softmax}\left(FC_2\left(FC_1\left(\text{AvgPool}\left(F_{ms}\right)\right)\right)\right) \\ F_c &= W_c \otimes F_{ms} \end{aligned} \tag{5}$$

$$\begin{aligned} W_s &= \text{Conv}_{5\times5}\left(\text{AvgPool}\left(F_c\right) \oplus \text{MaxPool}\left(F_c\right)\right) \\ F_g &= \mathbb{C}\left(W^i_s \otimes F_c\right). \end{aligned} \tag{6}$$

In (5), $FC_1(\cdot)$ and $FC_2(\cdot)$ denote two fully connected layers that produce a weight map W_c after applying the softmax activation function Softmax($\cdot$). The weight map W_c is subsequently used to modulate the input feature map F_{ms} via elementwise multiplication, resulting in the weighted feature map F_c.

In (6), the results of Avgpool($\cdot$) and Maxpool($\cdot$) operations are combined via an elementwise addition operator $\oplus$, and then processed through a convolutional layer with 5×5 kernel, denoted as $\text{Conv}_{5\times5}(\cdot)$. This convolution produces a set of spatial weight maps W_s^i for each landmark i. Each W_s^i is used to reweight F_c through elementwise multiplication. The reweighted features from all landmarks are then concatenated to form the final output feature map.

D. Training the Network

We began by training the alignment network independently. After completing its training, the alignment network was integrated as a fixed module, enabling us to advance to train the coarse-to-fine framework.

During the training of the alignment network, the ground-truth alignment box for each CT/CBCT scan was generated by expanding a predefined proportion from the smallest bounding box that encloses all landmarks. The alignment loss is defined in the following equation:

$$L_{\text{align}} = 1/6 \times \sum \sqrt{\left(\hat{B} - B\right)^2} \tag{7}$$

where $B - [x_{\text{begin}}, y_{\text{begin}}, z_{\text{begin}}, w, d, h]$ denotes the bounding box vector comprising the starting landmark coordinates and length, width, and height of the ground-truth alignment box. $\hat{B}$ represents the predicted bounding box.

During the extraction of landmarks from Gaussian heatmaps, the influence of distant values decreases significantly. To address this, we emphasize learning the heat values in the vicinity of the landmarks by generating masked heatmaps. Specifically, the heat values within a predefined radius around each landmark are doubled, while values outside this radius are set to -1. The computation of the Gaussian heatmap with mask for the ith landmark is formulated in the following equation:

$$\begin{aligned} H^i = 2 \times \exp\left(-\frac{(x-x_i)^2 + (y-y_i)^2 + (z-z_i)^2}{0.2 r_{\text{focus}}^2}\right) \times M^i \\ + M^i - 1 \end{aligned} \tag{8}$$

where r_{focus} denotes the focus radius, and M represents the focus mask with the same spatial dimensions as H. Within the radius of each landmark, the value of M is set to 1, whereas it is set to 0 beyond this radius. x, y, z represent the coordinate values of each voxel in heatmap H^i, and x_i, y_i, z_i represent the ground-truth coordinate values of the ith landmark.

In the training phase of the coarse-to-fine framework, we utilize a heatmap regression loss with mask [36] which comprises two components: heatmap focus loss and mask loss, as formulated in (10). Specifically, the heatmap focus loss is designed to capture the positional information of the landmarks, while the mask loss focuses on learning background information to reduce interference. Here, the commonly used Smooth $L1$ loss as shown in (9) for landmark detection is adopted to form the heatmap focus loss and the mask loss as shown in (10)

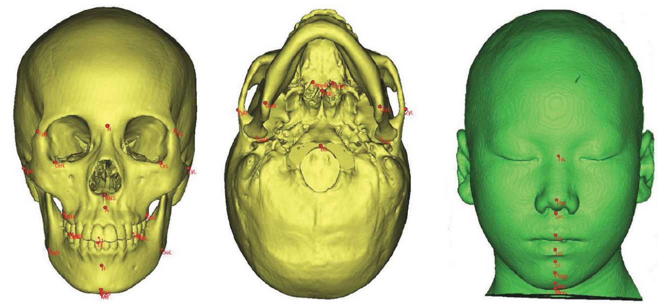

Fig. 5. Visualization of landmarks in a CT scan.

$$L_{\text{Smooth}L1}(\hat{y}, y) = \begin{cases} 0.5 * (\hat{y} - y)^2 & \text{, if } |\hat{y} - y| < 1 \\ |\hat{y} - y| - 0.5, & \text{otherwise.} \end{cases} \tag{9}$$

$$\begin{aligned} L_{\text{heatmap}} = L_{\text{Smooth}L1}\left(\hat{H} \times \rho_{\text{focus}}, H \times \rho_{\text{focus}}\right) \\ + L_{\text{Smooth}L1}\left(\hat{H} \times \rho_{\text{mask}}, H \times (1 - \rho_{\text{focus}})\right). \end{aligned} \tag{10}$$

Among them, ρ_{focus} is a binary matrix with the same dimensions as H, where the corresponding elements of H with heating value not less than 0 are set to 1, and the rest of the elements are set to 0. The top n_{mask} values in $\hat{H}$ are set to 1 in ρ_{mask}, and the rest are set to 0. n_{mask} denotes the count of 0 in the ρ_{focus}.

To balance the learning process between the global and local stages during training, we adopt an epoch-dependent training strategy. Specifically, the network in global stage is exclusively trained for the first 150 epochs. Then, both stages are trained simultaneously. Since the output of the global stage provides a reference for the input of the local stage and directly influences the local patch cropping, inaccurate global detection results can significantly impair the performance of the local network. The additional 150 epochs of global-stage training ensure stable and reliable global detection results, thereby positively contributing to the training of the local stage. The overall loss function is formulated as the weighted sum of the losses of the two stages, i.e., L_{global} and L_{local}, which is depicted in (11), where the weighting factor α dynamically adjusts to reflect the training progression. Both L_{global} and L_{local} are calculated according to (10)

$$L = \alpha L_{\text{global}} + L_{\text{local}}, \alpha = \max\left(0.1, 2.5 - 0.01 \times \text{epoch}\right). \tag{11}$$

Here, α is designed to gradually decrease from an initial value of 1 to a minimum of 0.1 over the first 150 epochs, ensuring a smooth transition from global to joint training while maintaining a significant contribution from the global loss throughout.

IV. Dataset and Experimental Setup

A. Datasets

To evaluate the proposed method, we constructed an in-house multimodality dataset comprising both CT and CBCT

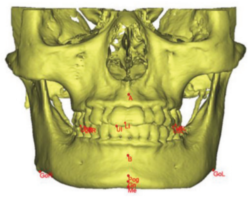

Fig. 6. Visualization of landmarks in a CBCT scan.

scans. The in-house CT dataset comprises 200 cranio-jaw-facial CT scans in DICOM format, with resolutions ranging from $256 \times 256 \times 196$ to $512 \times 666 \times 279$. The data were collected from the imaging database of the West China School/Hospital of Stomatology, Sichuan University, covering patients with permanent dentition aged between 16 and 60 years from September 2020 to March 2022, with an average age of 29.7 ± 8.41 years. The CT scans were acquired using a Philips MX 16-slice scanner (Philips, Dongqin) with the following scanning parameters: 120 kV tube voltage, 230 mAs tube current, 0.5 mm slice thickness, and 0.5 mm interslice spacing. No additional radiation was incurred for the patients, and all collected data were deidentified to remove personal information. Each sample was annotated by two senior orthodontists, one with 9 years and the other with 14 years of experience, and subsequently verified by a chief orthodontist with 31 years of experience. A total of 40 manually annotated landmarks were identified on the skull for each sample, as illustrated in Fig. 5. Notably, some samples exhibit incomplete annotations due to tissue deficiencies. The dataset is divided into 140 samples for training, 20 for validation, and 40 for testing.

The in-house 3-D CBCT dataset comprises 287 cranio-jaw-facial CBCT scans in DICOM format, with resolutions ranging from $256 \times 256 \times 141$ to $565 \times 405 \times 256$. The data were collected from March 2022 to October 2022 for patients aged between 16 and 60 years, with an average age of 33.2 ± 7.09 years. The CBCT scans were obtained using a Morita 3-D Accuitomo scanner (J. Morita, Japan) with scanning parameters of 85 kV tube voltage, 5 mA tube current, a scanning duration of 17.5 s, and a voxel size of 0.25 mm. This dataset includes 105 normal cases and 182 cases exhibiting malocclusion. Additionally, it shares the same data source and annotators as the in-house CT dataset. The average resolution of the CBCT scans is $561 \times 403 \times 143$, with each scan annotated with 13 landmarks, as shown in Fig. 6. The dataset is divided into 200 samples for training, 27 for validation, and 60 for testing.

To evaluate interannotator agreement and validate consistency, we calculated the intraclass correlation coefficient (ICC) between two orthodontists. The ICC is a robust metric for assessing the degree of agreement among multiple annotators. It captures both consistency (relative agreement in ranking) and absolute agreement (similarity in absolute values) across annotators. The ICC is derived from variance components analysis, which serves as a reliable indicator of annotation quality, with higher values suggesting greater reproducibility and reduced susceptibility to subjective biases. In our analysis, the ICC values between the two orthodontists were 0.917 for the CT dataset and 0.922 for the CBCT dataset, demonstrating strong agreement and high annotation reliability.

The ethical review and approval for the dataset used in this study was granted by the Ethics Committee of the West China School/Hospital of Stomatology, Sichuan University (Approval WCHSTRB-D-2023-011).

B. Training Details

The model was trained with an Intel Core i7-13700 CPU and four 12 GB GeForce GTX Titan X GPUs. During the alignment stage, the CT/CBCT images are rescaled to a uniform resolution of $24 \times 24 \times 24$ before being input into the alignment network. Alignment boxes are dynamically generated by expanding the minimal bounding box encompassing all landmarks by a factor of 0.2 for CT and 0.3 for CBCT. To enhance robustness, during training, a random expansion ratio within the range of $[-0.05, 0.05]$ is applied before cropping the alignment boxes. During testing, cropping is performed directly using the predicted alignment boxes. After alignment, the cropped images are uniformly resized to $128 \times 128 \times 64$ for the global stage. For the local stage, patches are cropped at a resolution of $32 \times 32 \times 16$ from the original images, centered on the landmarks. However, for landmarks near the image boundaries, the corresponding patches are constrained to remain within the original image dimensions. Regarding the generation of ground-truth heatmaps, the focus radius is set to 15 for coarse detection and 5 for fine detection, reflecting the different resolution requirements.

The alignment network is first trained independently in the first stage and subsequently used as a fixed computational component in subsequent stages. The weights of 3-D U-Net are randomly initialized, and a consistent four-layer architecture is employed throughout. Both the global and local stages are trained with a batch size of 4, where the contributions of the local loss are accumulated to compute the total loss for each iteration. The global and local stages of the coarse-to-fine framework are trained as a unified network for 400 epochs. An initial learning rate of 0.001 is used for CT and 0.0001 for CBCT, with the Adam optimizer and a weight decay coefficient of 0.0005 for parameter updating. An exponential decay strategy with a decay rate of 0.98 is applied, gradually reducing the learning rate to 0.003 of its initial value. The model demonstrating the best performance on the validation set during training is selected for final evaluation on the test set.

C. Evaluation Metrics

To evaluate the accuracy of the proposed method, three metrics were utilized: mean radial error (MRE, in mm), standard deviation (SD, in mm), and success detection rate (SDR, in %). The radial error R, as defined in (12) represents the Euclidean distance between the predicted and ground-truth landmarks, where Δx, Δy, and Δz are the differences along the x-, y- and z-axes, respectively

$$R = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}. \tag{12}$$

The MRE and SD are defined in the following equations, where N represents the total number of samples

$$\mathrm{MRE} = \frac{\sum_{i=1}^{N} R_i}{N} \tag{13}$$

$$\mathrm{SD} = \sqrt{\frac{\sum_{i=1}^{N} (R_i - \mathrm{MRE})^2}{N-1}}. \tag{14}$$

SDR denotes the proportion of predicted landmarks whose radial error falls below a specified threshold. The SDR was evaluated at target radii of 2.0, 2.5, 3.0, and 4.0 mm, respectively.

V. Experimental Results and Discussion

In this section, we first compare our proposed method with state-of-the-art approaches. Subsequently, ablation studies and analyses are conducted to demonstrate the effectiveness of the proposed modules. Additionally, external validation is performed on the publicly available oral MMLD dataset [26]. Furthermore, we explore the differences in landmark detection outcomes between CT and CBCT scans and evaluate the critical factors influencing the selection of the optimal imaging modality for clinical applications. We also investigate the role of different multiscale information extraction modules and attention modules, as well as examine the impact of varying patch sizes in the local stage on the final detection performance. Finally, we demonstrate the model's applicability in clinical settings by comparing its detection results with those of expert and discuss its transfer learning capabilities.

A. Comparison With SOTA Methods

To demonstrate the effectiveness of the proposed method, we conducted a comprehensive comparison with several state-of-the-art approaches [8], [9], [18], [26], [27], [39], [40]. The overall detection results are summarized in Table I. The detection performance was evaluated using MRE, SD, and SDR on both the CT and CBCT datasets.

The proposed method outperforms all other approaches across all evaluation metrics, as shown in Table I. Specifically, our method achieves the lowest MRE of 1.40 $\pm$ 0.47 mm on the CT dataset and 1.20 $\pm$ 0.32 mm on the CBCT dataset. In terms of SDR, our method consistently outperforms the other approaches, achieving 81.60%, 88.15%, 92.19%, and 95.78% across target radii of 2, 2.5, 3, and 4 mm, respectively, on the CT dataset. On the CBCT dataset, it achieves an 87.17% SDR for a 2 mm target radius.

Among these comparative methods, VNet [39], SCN [18], and ABRM [26] are single-stage methods, while PWA [8], SA-LSTM [9], LA-GCN [40], and DRM [27] are multistage methods. As shown in Table I, multistage methods generally outperform single-stage methods, due to their ability to iteratively refine landmarks, resulting in higher accuracy. In detail, VNet employs Gaussian heatmap regression for landmark detection and serves as a basic single-stage detection method. Although its performance is respectable for a single-stage model, it lags behind more advanced approaches, achieving an MRE of 2.49 $\pm$ 0.50 mm on CT dataset and 1.85 $\pm$ 0.30 mm on CBCT dataset. SCN is a classic 2-D single-stage network, which leverages spatial configuration to resolve ambiguities in Gaussian heatmaps. It achieves comparable accuracy, with an MRE of 2.16 $\pm$ 0.49 mm on CT dataset and 1.95 $\pm$ 0.55 mm on CBCT dataset. While SCN and VNet are anticipated to demonstrate comparable performance given their methodological designs, the CT dataset's larger number of landmarks underscores SCN's disambiguation capability, resulting in better performance compared to VNet on the CT dataset. Conversely, SCN shows inferior performance on the CBCT dataset. ABRM, an object detection method based on YOLO, demonstrates improved results compared to other single-stage methods. It achieves an MRE of 1.24 $\pm$ 0.28 mm on the CBCT dataset. However, its improvement on the CT dataset is less pronounced, with an MRE of 1.54 $\pm$ 0.49 mm, primarily because the CT dataset contains more landmarks than the CBCT dataset. ABRM performs less effectively when landmarks are more scattered and numerous. While the proposed method outperforms in terms of MRE and 2 mm SDR, it exhibits lower performance in SD and SDR at larger target radii. This weaker performance in detecting landmarks with larger errors is primarily attributed to the decline in detection accuracy when landmarks in local-stage patches are located near the edges or fall outside the patch boundaries.

PWA is a 2-D coordinate regression network based on ResNet. For comparison, we implemented a 3-D version. However, the attention mechanism of the network struggles to accurately map 3-D images to coordinates, resulting in suboptimal performance. PWA achieves an MRE of 3.70 $\pm$ 1.13 mm on the CT dataset and 3.30 $\pm$ 0.88 mm on the CBCT dataset. DRM employs a two-stage U-Net architecture with expansive exploration to enhance local detection capabilities. Our implementation extends DRM into a 3-D version, yielding competitive performance. DRM achieves an MRE of 1.67 $\pm$ 0.48 mm on CT dataset and 1.33 $\pm$ 0.40 mm on the CBCT dataset. LA-GCN is another two-stage heatmap regression method based on 3-D U-Net architecture. It incorporates graph convolution for multitask learning to determine the presence of landmarks, thereby addressing the issue of missing annotations. Additionally, it uses attention encoding of global structural information to detect landmarks and assist the secondary task. LA-GCN achieves an MRE of 1.73 $\pm$ 0.51 mm on the CT dataset and 1.35 $\pm$ 0.35 mm on the CBCT dataset. SA-LSTM incorporates LSTM to combine global structural information for refining landmarks. However, it struggles with mapping 3-D images to coordinates, resulting in lower detection accuracy. On the CT dataset, its MRE is 2.36 $\pm$ 0.61 mm, while on the CBCT dataset, it achieves an MRE of 2.07 $\pm$ 0.49 mm. In summary, our multistage landmark detection framework, combined with the global–local integration strategy, achieves optimal performance. We further analyze the effectiveness of the proposed modules in Sections V-B–V-D.

Regarding the parameters and inference time of the compared methods, as shown in Table I, the inference time of the single-stage object detection method ABRM is the lowest, but its detection accuracy is slightly lower than our method. The computational overhead of our method is primarily attributable

TABLE III
ABLATION EXPERIMENTS ON CBCT DATASET BY EVALUATION METRICS OF PARAMS, TIMES, MRE, SD, AND SDR IN FOUR TARGET RADII

Config	Params (M)	Times (ms)	Stage	MRE±SD (mm)	SDR (%)			
					2 mm	2.5 mm	3 mm	4 mm
Base	11.95	8249	Global	1.46±0.40	78.44	87.43	90.44	95.96
			Final	1.30±0.39	85.60	91.25	93.00	95.83
Base + GIEM	12.12	8280	Global	1.49±0.45	80.56	87.96	92.86	96.30
			Final	1.25±0.42	86.64	90.87	93.52	96.16
Base + MSGIEM	12.22	8818	Global	1.54±0.32	74.34	84.66	91.67	96.16
			Final	1.22±**0.30**	**87.43**	92.06	**94.18**	96.83
Base + MSGIEM + LAM	12.23	9142	Global	1.45±0.35	80.29	88.36	93.78	**97.75**
			Final	**1.20**±0.32	87.17	**92.72**	**94.18**	96.43

to its multistage strategy, i.e., the local stage performs dedicated landmark detection for each patch, which operates in a serial computation. But this could be addressed to a certain degree by group parallel inference and multiple GPUs support. Among the multistage methods, the second stage of SA-LSTM employs a coordinate regression model, resulting in relatively faster inference. However, due to the introduction of redundant information and the limitations of the coordinate regression method, its accuracy is suboptimal. Both LA-GCN and our proposed method utilize a two-stage Gaussian heatmap regression strategy, leading to similar inference times. However, the incorporation of residual convolution and graph convolutional branches in LA-GCN results in the highest number of parameters. In contrast, our method employs a lightweight U-Net structure with the fewest parameters. In summary, although this two-stage coarse-to-fine designation sacrifices computational efficiency, it achieves superior detection accuracy. Furthermore, real-time inference is not a strict requirement in clinical practice, and the inference time of our method remains within an acceptable range.

B. Ablation Study

To validate the effectiveness of various modules in our proposed method, we conducted a series of experiments on the CT and CBCT datasets respectively, with results presented in Tables II and III. The baseline method (denoted as Base in Tables II and III) consists of our three-stage framework, which includes the alignment network and the two-stage 3-D U-Net for landmark detection without incorporating any global features. Subsequently, the performances of the proposed MSGIEM and the LAM are evaluated separately.

To further evaluate the effectiveness of embedding global information into the local stage, we design a comparative experiment by introducing a basic Global Information Extraction Module (GIEM). This module directly merges the lowest-level feature maps from the global stage into the local stage. Additionally, the final results, along with the results obtained solely from the global stage (denoted as "Global") are presented in the Tables II and III, to highlight the effectiveness of the proposed modules and the local stage.

1) Effect of GIEM/MSGIEM: As shown in the first two rows of Tables II and III, embedding global information into the local stage significantly improves performance over the baseline, with a reduction in MRE of 0.05 mm on both the CT and CBCT datasets.

Moreover, the proposed MSGIEM yields further improvements compared to the baseline (Baseline + MSGIEM versus Baseline). As shown in Table II, the final MRE decreases by 0.13 mm, and the SDR is further improved by 0.61%, 0.53%, 0.31%, and 0.69% at target radii of 2, 2.5, 3, and 4 mm, respectively, on the CT dataset. On the CBCT dataset, as shown in Table III, the final MRE decreases by 0.08 mm, and the SDR improves by 1.83% at target radii of 2 mm. Compared to GIEM, the proposed MSGIEM demonstrates superior performance. As shown in Table II, incorporating MSGIEM into the baseline model (Baseline + MSGIEM versus Baseline + GIEM) results in a 0.08 mm reduction in MRE on the CT dataset, along with notable SDR improvements at the 3 mm and 4 mm thresholds.

Similarly, as shown in Table III, the CBCT dataset exhibits a 0.03 mm decrease on MRE and SDR improvements of 0.79%, 1.19%, 0.66%, and 0.67% at target radii of 2, 2.5, 3, and 4 mm, respectively, compared to GIEM. These improvements indicate that MSGIEM enhances detection performance by leveraging pyramid convolutions to capture global features at multiple scales. This allows different landmarks to benefit from varying levels of global information, enabling more accurate and robust landmark detection.

2) Effect of LAM: As shown in Table II, the inclusion of the LAM reduces the final MRE by 0.09 mm, and improves the SDR by 2.46%, 1.13%, and 0.17% at target radii of 2, 2.5, and 3 mm, respectively, on the CT dataset. On the CBCT dataset, as shown in Table III, the final MRE decreases by 0.02 mm. These results demonstrate that the LAM effectively enhances the expression of global information, leading to improved detection accuracy. This improvement stems from the fact that different landmarks are distributed across distinct regions, each focusing on different semantic levels and spatial information. The proposed LAM amplifies relevant global information for each landmark while suppressing irrelevant information, thereby achieving better results. Here, for the experiments involving the baseline and baseline with GIEM/MSGIM, the same global features were applied to local detections for different landmarks. Although the improvement on the CBCT dataset is less pronounced, this is primarily due to the smaller number of landmarks in the CBCT dataset compared to the CT dataset. The effectiveness of the LAM is more significant when applied to datasets with a larger number of landmarks.

Furthermore, in terms of inference time, the introduction of MSGIEM and LAM modules resulted in an increase of

TABLE IV
VALIDATION ON PUBLIC MMLD DATASET BY EVALUATION METRICS OF MRE, SD, AND SDR IN FOUR TARGET RADII

Method	MRE±SD (mm)	SDR (%)			
		2 mm	2.5 mm	3 mm	4 mm
UNet3D	2.54±0.83	45.29	59.59	70.91	86.36
Two-Stage [36]	1.78±0.81	68.93	82.48	88.68	95.37
ABRM [26]	1.66±0.92	79.26	88.43	92.64	**96.78**
Proposed Method	**1.52±0.61**	**80.86**	**88.82**	**92.90**	95.91

approximately 1 s in inference time on both datasets. This indicates that these two modules have a certain impact on inference speed, but they account for a relatively small proportion of the total inference time. To improve inference speed, the inference time in the local stage is worth further investigation in future work by incorporating parallel strategy and more hardware support. Subsequently, the results of the global stage, as shown in Tables II and III, reveal that the MRE exceeds 2 mm on the CT dataset and 1.4 mm on the CBCT dataset. This discrepancy is primarily due to the fact that CBCT images focus on the oral and maxillofacial region, containing less redundant information, whereas the CT dataset includes more challenging landmarks to be detected. On both datasets, the detection results in the local stage show significant improvements compared to the global stage, demonstrating the effectiveness of the multistage approach. Additionally, since the global and local stages are trained as a unified model, the loss from the local stage influences the weight parameters of the global stage encoder, leading to instability of the global stage results. This instability is particularly pronounced after incorporating the MSGIEM, which causes a notable drop in accuracy compared to the baseline. However, the results of the local stage are improved, indicating that enhancing the local stage has a more significant effect on the overall detection performance.

C. Evaluation on Public Dataset

To validate the robustness of our method, we conducted experiments on the publicly available mandibular molar landmark dataset (MMLD) [26]. The MMLD dataset, sourced from West China School/Hospital of Stomatology, Sichuan University, comprises 658 CT scans, with 458 for training, 100 for validation, and 100 for testing. It is noted that the complete CT scans are given in the training and validation sets, while the test set provides cropped CT region scans. Due to the inconsistency of the test set data, we used the training set for training and the validation set for testing to ensure the robustness of the method. The results are presented in Table IV, which demonstrate that our method outperforms state-of-the-art approaches, achieving accurate localization on the mandibular molars. The MRE is 1.52 mm, with SDR of 80.86%, 88.82%, 92.90%, and 95.91% at target radii of 2, 2.5, 3, and 4 mm, respectively. While the 4 mm SDR is slightly lower compared to the ABRM, the MRE improved by 0.26 mm, and the 2 mm SDR increased by 11.93% compared to the basic two-stage network.

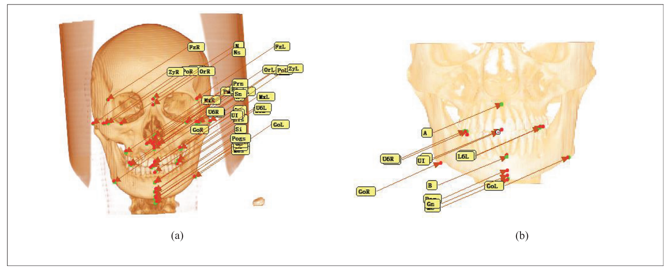

Fig. 7. Visualization of landmarks on CT and CBCT dataset. The green dots denote ground-truth landmarks, and the red dots denote predicted landmarks.

TABLE V
COMPARISON OF LANDMARKS ON BOTH CT AND CBCT DATASETS BY EVALUATION METRICS OF MRE AND SD

Landmark	MRE±SD (mm)	
	CT	CBCT
A	1.22 ± 0.86	1.41 ± 1.33
B	1.56 ± 1.05	1.30 ± 0.92
Gn	1.02 ± 0.51	0.91 ± 0.48
GoL	2.20 ± 1.68	1.85 ± 1.22
GoR	2.13 ± 1.51	1.85 ± 1.62
L6L	1.25 ± 1.35	1.14 ± 1.09
L6R	1.13 ± 0.48	1.37 ± 1.38
LI	1.41 ± 1.36	1.16 ± 0.97
Me	1.36 ± 0.85	1.02 ± 0.56
Pog	1.04 ± 0.45	0.86 ± 0.50
U6L	1.23 ± 1.66	0.98 ± 1.13
U6R	1.21 ± 1.08	1.03 ± 0.96
UI	0.78 ± 0.54	0.77 ± 0.76
Mean	1.33 ± 0.78	1.20 ± 0.32

D. Summary and Discussion

To investigate the effectiveness of deep learning-based landmark detection methods across different data modalities, we evaluated the detection performance of common landmarks present in both the CT and CBCT datasets. Subsequently, we compared various multiscale information extraction modules and attention modules to illustrate the effectiveness of our proposed MSGIEM and LAM. Additionally, we examined the impact of different patch sizes in the local stage on detection results. Finally, we validated the model's practical utility in clinical environments by comparing its detection outcomes with those of medical experts and explore its potential for transfer learning.

TABLE VI
COMPARATIVE EXPERIMENTS OF DIFFERENT MULTISCALE INFORMATION EXTRACTION MODULE ON CT AND CBCT DATASETS BY EVALUATION METRICS OF MRE, SD, AND SDR IN FOUR TARGET RADII

Config	CT Dataset					CBCT Dataset				
	MRE±SD (mm)	SDR (%)				MRE±SD (mm)	SDR (%)			
		2 mm	2.5 mm	3 mm	4 mm		2 mm	2.5 mm	3 mm	4 mm
SFB [41]	1.46±0.48	79.65	87.40	91.75	95.34	1.21±**0.29**	85.85	91.93	**94.18**	96.30
M2SM [42]	1.41±**0.44**	81.16	88.09	91.87	**96.41**	1.23±0.32	86.81	92.26	93.99	96.28
MSGIEM	**1.40**±0.47	**81.60**	**88.15**	**92.19**	95.78	**1.20**±0.32	**87.17**	**92.72**	**94.18**	**96.43**

TABLE VII
COMPARATIVE EXPERIMENTS OF DIFFERENT ATTENTION MODULE ON CT AND CBCT DATASETS BY EVALUATION METRICS OF MRE, SD, AND SDR IN FOUR TARGET RADII

Config	CT Dataset					CBCT Dataset				
	MRE±SD (mm)	SDR (%)				MRE±SD (mm)	SDR (%)			
		2 mm	2.5 mm	3 mm	4 mm		2 mm	2.5 mm	3 mm	4 mm
SAM	1.44±0.48	80.05	87.61	91.95	95.10	1.24±0.35	86.94	92.18	93.83	96.04
CAM	1.43±0.44	81.08	87.94	92.06	95.42	1.21±0.33	87.02	92.53	93.96	96.28
LAM	**1.40±0.47**	**81.60**	**88.15**	**92.19**	**95.78**	**1.20±0.32**	**87.17**	**92.72**	**94.18**	**96.43**

1) Discussion of Landmark Detection on CT and CBCT Datasets: In clinical practice, physicians typically choose between CT or CBCT imaging based on patient circumstances, radiation exposure, and cost considerations. CBCT scans are associated with lower radiation doses and costs, while CT scans provide a larger field of view, encompassing the upper face and neck. However, CT images generally exhibit relatively lower spatial resolution and greater slice thickness compared to CBCT. In our private datasets, the slice thickness for CT scans is 1 mm, whereas it is 0.25 mm for CBCT scans. To investigate the selection criteria for CBCT versus CT examinations and their impact on detection accuracy, we compared the performance of common landmarks across both CT and CBCT datasets. Fig. 7 illustrates the visualization results of our method for landmark detection on CT and CBCT datasets. The 13 annotated landmarks in the CBCT dataset are also annotated in the CT dataset. The quantitative results, presented in Table V, indicate that the detection accuracy on the CBCT dataset is generally higher than that on the CT dataset. This improvement may be attributed to the lower overall redundant information and smaller slice thickness of CBCT images. Furthermore, the accuracy for challenging landmarks, such as GoL and GoR, is below the average level in both datasets. The errors of these landmarks may be caused by individual anatomical variations. In contrast, landmarks with distinct features, such as UI, exhibit higher detection performance. This suggests that neither modalities show bias toward specific landmarks. Although landmark detection on CBCT images demonstrates high accuracy, its limited field of view may not fully meet the requirements of certain clinical surgical procedures. For automatic landmark detection focused solely on the oral and maxillofacial region, CBCT is recommended. However, if the neck or upper facial regions are required, CT is the preferred images modality.

2) Discussion of Different Multiscale Information Extraction Modules: To further validate the effectiveness of our proposed MSGIEM, we conducted comparative experiments with two state-of-the-art multiscale information extraction modules: spatial fusion block (SFB) [41] and multiscale-in-multiscale subtraction module (M2SM) [42]. The SFB was adopted from landmark detection due to its proven effectiveness in capturing spatial relationships and contextual information across different scales. The M2SM has demonstrated superior performance in handling scale variations through its hierarchical feature processing architecture. They represent the advanced techniques in respective domains. The experimental results are presented in Table VI. While SFB achieved competitive SD values (0.29 mm) on CBCT datasets through its spatial feature fusion strategy, it showed limitations in handling multilevel semantic information, particularly evidenced by its lower SDR values (85.85% at 2 mm) compared to our method. M2SM demonstrated strong performance in 4 mm SDR (96.41%) on CT datasets through its hierarchical feature subtraction design but exhibited instability in cross-modality scenarios with lower SDR values on CBCT datasets.

MSGIEM module addresses these limitations through its pyramid-based multiscale fusion strategy. By systematically processing feature maps from four resolution levels in the global-stage encoder, MSGIEM achieves unified channel representation (64 channels) and spatial resolution ($32 \times 32 \times 16$) through convolution, upsampling, and pooling operations. This standardized processing enables effective aggregation of anatomical structural information across different semantic levels through pyramidal convolution. The experimental results demonstrate that MSGIEM outperforms both modules in 4 of the 6 key metrics on CT datasets and 5 of the 6 metrics on CBCT datasets, particularly showing significant improvements in challenging 2 and 2.5 mm SDR thresholds.

3) Discussion of Different Attention Module: To demonstrate the effectiveness of our proposed LAM, we conducted comparative experiments by replacing LAM with a spatial attention module (SAM) and a channel attention module (CAM). The results of these experiments are summarized in Table VII. In these experiments, the global information extracted by SAM and CAM is equally fused into the landmark detection process in the local stage. CAM produced better

TABLE VIII

COMPARATIVE EXPERIMENTS OF DIFFERENT PATCH SIZE ON CT AND CBCT DATASETS BY EVALUATION METRICS OF MRE, SD, AND SDR IN FOUR TARGET RADII

Config	CT Dataset					CBCT Dataset				
	MRE±SD (mm)	SDR (%)				MRE±SD (mm)	SDR (%)			
		2 mm	2.5 mm	3 mm	4 mm		2 mm	2.5 mm	3 mm	4 mm
16 × 16 × 8	1.48±0.57	79.21	87.40	91.24	95.65	1.27±0.39	86.17	91.95	93.46	96.34
32 × 32 × 16	**1.40±0.47**	**81.60**	**88.15**	**92.19**	**95.78**	**1.20±0.32**	**87.17**	**92.72**	**94.18**	**96.43**

TABLE IX

COMPARATIVE EXPERIMENTS OF MODEL AND EXPERT ON CT AND CBCT DATASETS BY EVALUATION METRICS OF TIMES, MRE, SD, AND SDR IN FOUR TARGET RADII

Config	CT Dataset						CBCT Dataset					
	Times (s)	MRE±SD (mm)	SDR (%)				Times (s)	MRE±SD (mm)	SDR (%)			
			2 mm	2.5 mm	3 mm	4 mm			2 mm	2.5 mm	3 mm	4 mm
Model	**12.6**	1.29±0.25	84.09	89.14	92.93	96.21	**9.1**	1.18±0.41	84.92	**92.06**	**95.24**	96.83
Expert	900	**1.24±0.14**	**85.35**	**90.66**	**95.20**	**97.47**	360	**1.08±0.27**	**89.68**	**92.06**	**95.24**	**98.41**

overall results than SAM, primarily due to its ability to capture multiscale global features. By selecting features across multiple scales, CAM enables more effective utilization of global information.

However, the LAM module outperforms both SAM and CAM by extracting unique global information tailored for each local detection network. The experimental results demonstrate that by focusing on multiscale channels and spatial attention, LAM can extract relevant global features for different landmark detections at the local stage. This approach effectively enhances landmark detection and achieves higher overall accuracy.

4) Discussion of Different Patch Sizes: To investigate the impact of different patch sizes and global feature inputs on the detection performance in the local stage, we designed comparative experiments with varying patch sizes. Specifically, we examine the use of a smaller patch size (16 × 16 × 8), corresponding to approximately 8×8×8 mm, in the local stage to minimize the influence of edge information. Additionally, we apply a pooling operation to the global features extracted by LAM to match the smaller patch size. Due to computational resource constraints, the largest patch size used is 32×32×16. Table VIII shows the comparative results of the two patch sizes on the CT and CBCT datasets. In both datasets, the 32×32×16 patches outperformed the smaller patches. This suggests that, within a certain range, a larger field of view provides more contextual information, aiding in landmark detection and helping to capture landmarks with larger localization errors from the global stage, thereby improving overall detection accuracy. And it is obvious that reducing the patch size leads to a significant decline in detection performance. In the future work, a high-performance GPU will be essential to test large-size patches.

5) Discussion of Clinical Applicability: To evaluate the clinical applicability of our method, we conducted a rigorous comparative analysis between our model's output and expert annotations on ten randomly selected cases from both CT and CBCT test sets, respectively. To further assess potential improvements, we enlisted a highly experienced expert with 14 years of expertise in this domain, who had previously been involved in similar annotation tasks two years ago, to perform a secondary annotation of 20 selected cases. As shown in Table IX, human expert achieves superior quantitative metrics with 1.24 mm MRE on CT and 1.08 mm on CBCT, slightly higher than the model's performance. Specifically, as shown in Tables X and XI, we presented the results of model and expert's performance on 13 landmarks, the analysis reveals that the expert outperforms the model at more complex or ambiguous landmarks, such as Landmark Gn in CT (expert's 0.98 ± 0.50 versus model's 1.17 ± 0.49) and Landmark LI in CBCT (expert's 0.38 ± 0.17 versus model's 0.57 ± 0.33). This is likely due to the expert's deeper understanding of anatomical variability and their ability to interpret challenging structures that may not be well-represented in the model's training data. On the other hand, the model performs better than the expert at certain landmarks, particularly where anatomical structures are more standardized or less complex. For example, in the CT data, the model shows better precision at Landmark Me (1.08 ± 0.53 versus expert's 1.75 ± 0.75), and in the CBCT data, the model achieves a significantly lower MRE at Landmark A (1.71 ± 1.15) compared to the expert (3.71 ± 1.76). These results suggest that the model excels in consistency and efficiency, especially for high-contrast or well-defined landmarks. Overall, the model's strength lies in its uniform processing and lower variability, while the expert's advantage stems from their nuanced interpretation of complex or noisy anatomical features.

Furthermore, the proposed model demonstrates significant efficiency improvements over manual annotation while maintaining clinically acceptable and comparative accuracy. The automated system completes landmark detection in 12.6 s (CT) and 9.1 s (CBCT) per case, representing a 40–71 speedup compared to expert manual annotation. It significantly reduces orthodontists' workload. While for some rare complex cases, it allows efficient manual refinement of specific landmarks to achieve higher detection accuracy. This human-AI

TABLE X
COMPARATIVE EXPERIMENTS OF MODEL AND EXPERT ON 13 LANDMARKS OF CT DATASETS BY EVALUATION METRICS OF MRE AND SD

		A	**B**	**Gn**	**GoL**	**GoR**	**L6L**	**L6R**	**LI**	**Me**	**Pog**	**U6L**	**U6R**	**UI**
Model	MRE (mm)	**1.16**	1.35	1.17	1.79	**1.62**	**0.85**	**0.93**	1.16	**1.08**	**0.89**	0.89	1.06	0.50
	SD (mm)	**0.48**	1.22	**0.49**	0.91	**1.04**	**0.61**	**0.37**	1.22	**0.53**	**0.35**	0.57	1.03	0.25
Expert	MRE (mm)	1.27	**1.21**	**0.98**	**1.09**	1.81	1.16	1.03	**0.83**	1.75	1.04	**0.80**	**0.64**	**0.35**
	SD (mm)	1.08	**0.75**	0.50	**0.74**	1.66	0.72	0.38	**0.17**	0.75	0.44	**0.46**	**0.26**	**0.21**

TABLE XI
COMPARATIVE EXPERIMENTS OF MODEL AND EXPERT ON 13 LANDMARKS OF CBCT DATASETS BY EVALUATION METRICS OF MRE AND SD

		A	**B**	**Gn**	**GoL**	**GoR**	**L6L**	**L6R**	**LI**	**Me**	**Pog**	**U6L**	**U6R**	**UI**
Model	MRE (mm)	**1.71**	1.10	0.87	1.72	1.59	1.53	1.89	0.57	**1.04**	**0.78**	0.85	0.74	0.70
	SD (mm)	**1.15**	0.85	0.67	0.86	1.10	1.31	2.59	0.33	0.76	0.60	0.38	0.33	0.33
Expert	MRE (mm)	3.71	**1.05**	**0.72**	**1.06**	**0.94**	**1.05**	**1.04**	**0.38**	1.45	0.82	**0.62**	**0.62**	**0.39**
	SD (mm)	1.76	**0.62**	**0.36**	**0.75**	**0.56**	**0.50**	**0.60**	**0.17**	**0.45**	**0.38**	**0.28**	**0.34**	**0.23**

collaboration maintains diagnostic reliability while optimizing workflow efficiency, enabling orthodontists to prioritize critical clinical decisions over repetitive measurement tasks.

6) Discussion of Model Transfer Learning Ability: The proposed framework can be readily extended to other clinical applications through efficient transfer learning. By preserving the hierarchical feature extraction backbone and reconfiguring the alignment network or output layers to suit target data, the model retains its cascaded detection precision while meeting domain-specific requirements. For instance, when applied to CBCT scans with enlarged fields of view, clinicians can prioritize fine-tuning the alignment network without sacrificing the inherent coarse-to-fine detection capability. This modular adaptability ensures robust performance across diverse imaging protocols while minimizing retraining overhead.

VI. CONCLUSION

In this work, we introduced a novel three-stage deep learning model designed for accurate and efficient landmark detection in 3-D CT and CBCT scans. The proposed model overcomes the limitations of existing multistage approaches by incorporating an MSGIEM and an LAM, which enhance the integration of global and local features during detection. By combining coarse and fine detection stages, our method achieves high-precision landmark detection, even in the presence of complex anatomical structures.

Experimental results demonstrate that the proposed method significantly outperforms state-of-the-art techniques on both CT and CBCT datasets, achieving lower MRE and higher SDR across various radii. This improvement is attributed to the innovative integration of global information into the local detection stage, which enhances the accuracy and robustness of the landmark detection process. However, our method relies on Gaussian heatmap regression, which is sensitive to anisotropy. As image slice thickness increases, so does the detection error. In the future work, we plan to explore the combination of coordinate regression and Gaussian heatmap regression to improve localization accuracy for thick image slices. Additionally, we plan to leverage anatomical prior information to design an adaptive cropping strategy, aiming to optimize local patch extraction. Furthermore, we intend to increase the patch size of the local stage to further enhance performance, contingent upon advancements in hardware capabilities.

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